

SANTRA AI — NeuroX 1.0 Phase 1 Report

Project: SANTRA AI autonomously monitors competitors, reasons over live web intelligence, generates executive-ready insights, and safely executes approved business actions.

Track: NeuroX 1.0 — Autonomous B2B GTM Intelligence Agent

Team: Prompt Pirates

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1. Executive summary

SANTRA AI is an autonomous B2B go-to-market (GTM) intelligence agent for revenue operations, competitive intelligence, and strategy teams.

Instead of analysts manually checking competitor pricing pages, job boards, and SERP results every week, SANTRA accepts a plain-language monitor goal, reasons about intent, routes to the right web-evidence tools, detects material changes, synthesizes an executive brief, and queues CRM or webhook automation for **human approval** before anything executes.

SANTRA AI monitors competitors, collects live web evidence, drafts executive briefs, and routes approved actions to your revenue stack.

Phase 1 deliverable: Working agent loop with GTM Monitors, live evidence collection (Bright Data + Exa fallback), executive reports, human-in-the-loop approval queue, and gated webhook execution.

2. Problem statement

B2B GTM and competitive intelligence teams face a recurring operational bottleneck:

- **Fragmented research:** Pricing pages, hiring pages, product changelogs, and SERP rankings live across dozens of URLs with no single system of record.
- **Slow reaction time:** A rival's pricing tier change or enterprise sales hiring surge may sit unnoticed for days while reps and leadership operate on stale battlecards.
- **Manual synthesis:** Analysts spend 15–30+ hours per week copying snippets into slides, Slack threads, and CRM notes - work that does not scale with portfolio breadth.
- **Automation without guardrails:** Teams want CRM updates and workflow triggers, but blind automation on unverified intelligence creates compliance and reputational risk.

Who feels this pain: RevOps leaders, competitive intelligence analysts, product marketing, and GTM strategy teams at B2B SaaS companies monitoring 5–50 named rivals.

Cost of inaction: Missed pricing moves, delayed counter-positioning, inconsistent field messaging, and low-trust automation that leadership refuses to adopt.

3. Value proposition

SANTRA AI compresses the competitive monitoring loop from days to minutes while keeping humans in control of outbound actions.

Before SANTRA	With SANTRA
Manual URL checks and spreadsheet trackers	Autonomous monitors with scheduled and on-demand checks
Ad-hoc Google searches and stale screenshots	Routed collection via Bright Data SERP/Unlocker/MCP, with Exa fallback when primary APIs fail
Analyst-written briefs after the fact	AI synthesis: verdict, risks, opportunities, actions

Before SANTRA	With SANTRA
Zapier flows firing on unverified signals	HITL approval queue — CRM/webhook runs only after explicit approve
Siloed chat tools with no audit trail	Workspace history, agent activity log, and report snapshots

Measurable outcomes (target): - 70%+ reduction in time from signal detection to executive-ready brief - Single workspace for monitors, evidence, reports, and approved automations - Audit-friendly trail: what was proposed, who approved, what executed

4. Target users and use cases

Primary users

1. **Competitive intelligence analyst** — maintains battlecards and pricing matrices
2. **RevOps / GTM ops** — wires approved signals into HubSpot, Salesforce, or internal webhooks
3. **Product marketing** — tracks positioning and launch moves
4. **Sales leadership** — consumes executive briefs before QBRs and deal strategy sessions

Representative monitors (plain-language goals) - “Alert me when ApexAnalytics changes enterprise pricing tiers” - “Track competitor hiring in enterprise sales roles” - “Monitor SERP rankings for ‘AI revenue intelligence’ keywords” - “Detect new security/compliance pages on rival sites”

5. Agentic loop (core product behavior)

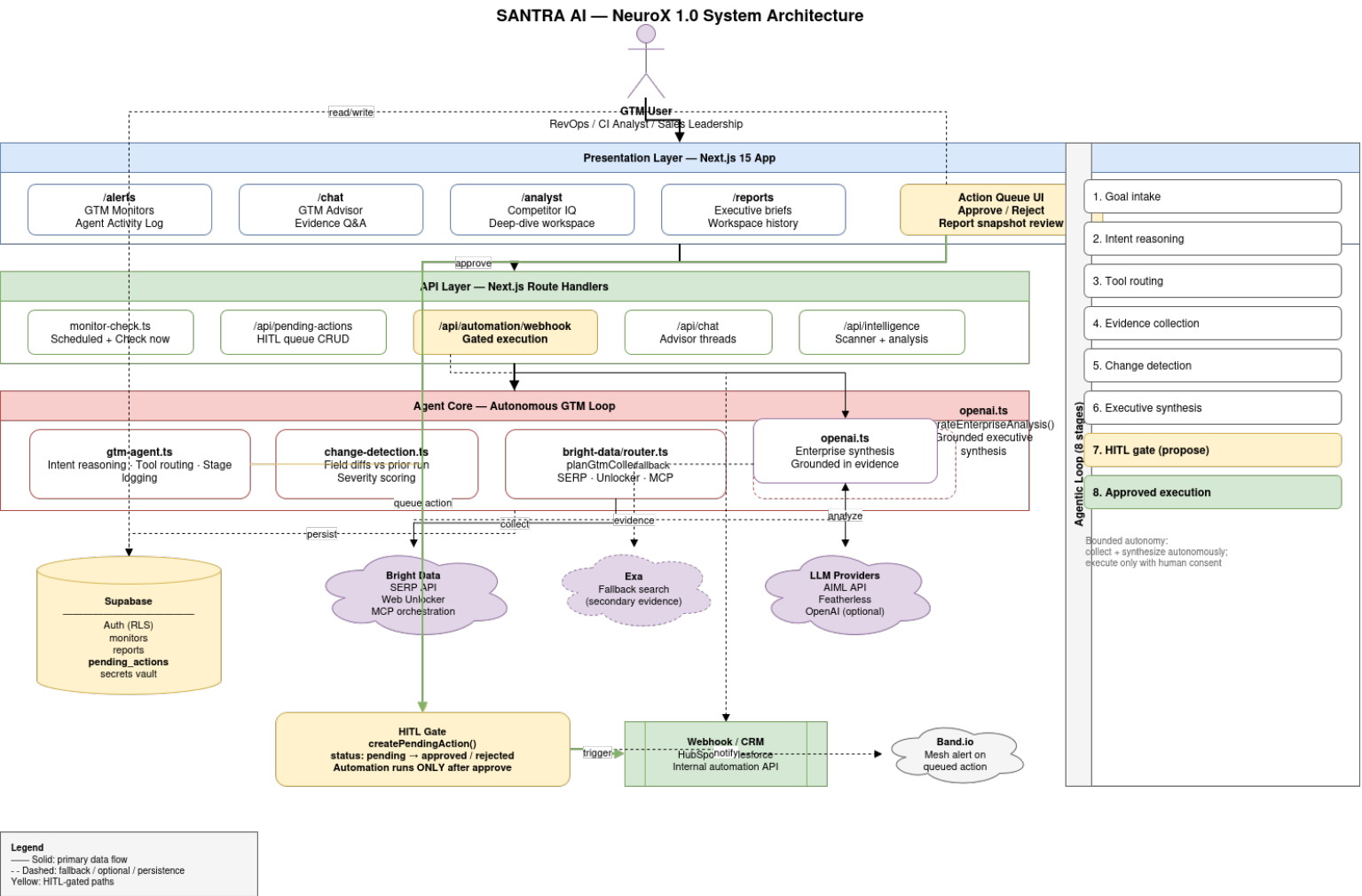
SANTRA implements a closed-loop autonomous agent with explicit stages logged in the UI:

1. Goal intake → User describes monitor in natural language
2. Intent reasoning → Category, severity, keywords, target URL inferred
3. Tool routing → Dynamic plan: SERP API, Web Unlocker, MCP, or Exa
4. Evidence collection → Live web snippets, structured change candidates
5. Change detection → Diff against prior run; severity scoring
6. Executive synthesis → Verdict, risks, opportunities, action plan
7. HITL gate → Pending action queued; Band.io optional notify
8. Approved execution → Webhook/CRM trigger only after human approve
9. Ambiguity → If the user's monitoring goal is unclear (e.g., "Track AI competitors"), SANTRA evaluates the workspace context, previously tracked competitors, and confidence scores. When confidence is low, it requests clarification before continuing.
10. Continuous Learning → Feedback from approved or rejected actions is stored to improve future recommendations and routing decisions.

Why this is agentic (not a chatbot): - The system **plans** which tools to call based on query shape and configuration. - It **acts** autonomously on schedule or "Check now" without re-prompting. - It **proposes** downstream automation but **waits** for approval — bounded autonomy aligned with enterprise GTM workflows.

6. System architecture

6.1 High-level diagram



6.2 Key modules

Module	Route / file	Responsibility
GTM Monitors	/alerts, monitor-check.ts	Orchestrates full agent loop per monitor
Agent core	gtm-agent.ts	Intent, routing, collection, stage logging
Evidence router	bright-data/router.ts	Plans SERP vs unlocker vs MCP steps
Exa fallback	exa-search.ts	Secondary web evidence when Bright Data unavailable
Analysis	openai.ts	Enterprise synthesis grounded in evidence
Change detection	change-detection.ts	Field-level diffs vs previous run
HITL queue	pending_actions table, /api/pending-actions	Stores proposed CRM/webhook actions
Automation gate	/api/automation/webhook	Executes only on approved actions
Notifications	band-agent.ts	Band.io mesh alert when action queued
GTM Advisor	/chat	Interactive competitor Q&A with evidence collection
Competitor IQ	/analyst	Deep-dive analyst workspace
History	/reports	Monitor reports and GTM briefings

6.3 Data model (HITL)

Migration 006_pending_actions.sql defines:

- proposed_action — human-readable summary (e.g. “Review and approve CRM trigger: Pricing tier change detected”)
- proposed_event — event type for webhook payload
- report_snapshot — frozen executive report at queue time
- status — pending → approved / rejected → execution

This design ensures automation never runs on stale or unreviewed intelligence.

7. Human-in-the-loop (HITL) design

Principle: Collect and synthesize autonomously; **execute** only with explicit human consent.

Flow:

1. Monitor check completes → executive report generated
2. If signals match monitor rules → createPendingAction() inserts row in Supabase
3. UI **Action Queue** panel shows proposed action + report snapshot
4. User clicks **Approve** → status updated → **Trigger workflow** calls gated webhook.
5. User clicks **Reject** → action archived; no outbound automation

Optional notification layer: Band.io agent receives a lightweight event when a new action is queued (requires configured chat room).

8. External tools and integrations

Tool	Role in SANTRA
Bright Data	Primary SERP, Web Unlocker, MCP evidence collection
Exa	Automatic fallback web search when Bright Data fails or credits exhausted
Supabase	Auth, monitors, reports, pending_actions, secrets vault
AIML API / Featherless	OpenAI-compatible LLM for analysis and chat
Band.io	Agent mesh notifications for HITL queue events
OpenAI	Alternate inference provider

9. Demo walkthrough

1. Open <https://santra-ai-neurox.vercel.app/alerts> (or localhost:3001/alerts)
2. **Create monitor:** “Monitor [Competitor] pricing page for tier changes” + target URL
3. Click **Check now** → watch **Agent Activity Log** (intake → routing → collection → synthesis)
4. Open generated **executive report** — verdict, risks, evidence preview
5. In **Action Queue**, review proposed CRM/webhook action
6. Click **Approve** → **Trigger workflow** → show webhook payload (or CRM export)
7. Optional: /chat — “Draft a battlecard vs [rival]” with live evidence

10. Reasoning and tool-routing design

Intent parsing: Monitor requirement text is enriched with workspace context (company, ICP, tracked rivals).

Dynamic routing (planGtmCollection): - URL-heavy goals → Web Unlocker / scraper path - Keyword/SERP goals → Bright Data SERP API - Complex multi-step → MCP orchestration when enabled

Resilience: - Reverse step order retry if first Bright Data pass returns demo/empty
- **Exa fallback** if Bright Data still unavailable - Stage log surfaces every decision for demo transparency

Grounded synthesis: generateEnterpriseAnalysis receives collected evidence — reduces hallucinated competitor claims.

Situation	Tool Selected
Search Engine Ranking	Bright Data SERP
Blocked Website	Bright Data Unlocker
API Failure	Exa Search
Historical Reports	Supabase
Executive Analysis	OpenAI / AIML

11. B2B impact

SANTRA targets a high-friction enterprise workflow — competitive intelligence — not consumer novelty.

- **Revenue relevance:** Pricing and positioning shifts directly affect win rates and pipeline quality.
- **Team scalability:** One analyst can monitor many rivals without linear headcount growth.
- **Adoption path:** HITL + CRM/webhook integration fits existing RevOps stacks instead of replacing them.
- Expected Business ROI
- 70–80% reduction in manual competitor research
- 5× faster executive report generation
- Monitor hundreds of competitors simultaneously
- Faster GTM response to pricing and positioning changes
- Reduced operational costs for RevOps teams

12. Architecture quality

- **Modular services:** Agent, research, analysis, change detection, and persistence are separated
- **Type-safe TypeScript** across API routes and UI
- **Supabase RLS** on user-scoped data
- **Graceful degradation:** Demo mode when keys missing; Exa when Bright Data exhausted
- **Test coverage:** API route tests under `src/app/api/**/__tests__`

13. Limitations and honest scope (Phase 1)

- Band.io notifications require an active agent chat room
- Bright Data credits may require Exa fallback during heavy demo usage
- CRM field mapping is webhook-generic; deep Salesforce object mapping is Phase 2

14. Phase 2 roadmap

1. **Richer approval UI** — edit proposed action text and CRM payload before execute
2. **Agent memory** — cross-run context (“rival raised prices twice in 90 days”)
3. **Slack-native approval cards** — approve/reject in-channel
4. **Deeper CRM mapping** — HubSpot/Salesforce field templates per monitor type
5. **Multi-monitor correlation** — “three rivals shifted enterprise positioning this month
6. Multi-monitor correlation — "Three rivals shifted enterprise positioning this month."

15. Why SANTRA is Agentic

- Unlike traditional AI assistants, SANTRA continuously plans tasks, selects tools based on context, retries failed operations, reasons over collected evidence, requests human approval for critical actions, and executes business workflows autonomously.
- The agent remains goal-driven rather than prompt-driven, making it suitable for enterprise GTM operations.

16. Team

Name	Role	Github
Ahamed AAH	Leader	@AhamedAAHA
Avashik Ahamed	Member	@kishava
Aathil Akmal	Member	@aathil-akmal

17. Appendix — repository and setup

git clone https://github.com/AhamedAAHA/Santra_AI_NeuroX

cd Santra-Live

npm install

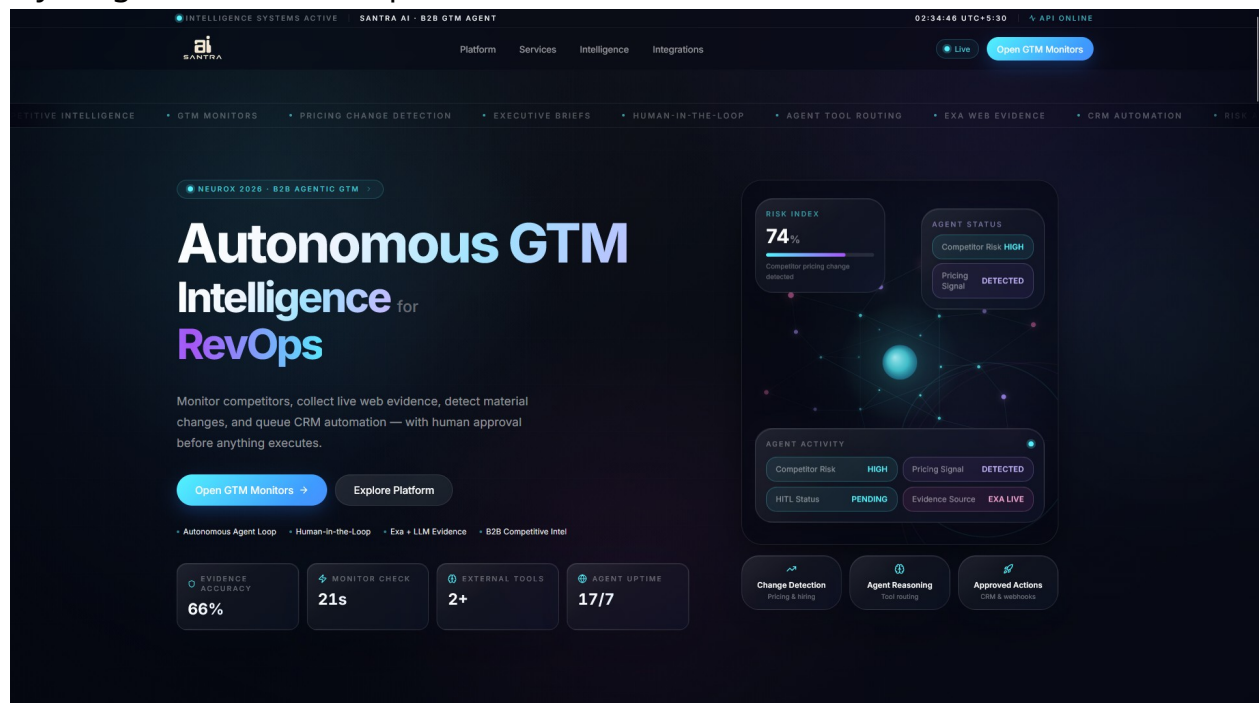
cp .env.example .env.local

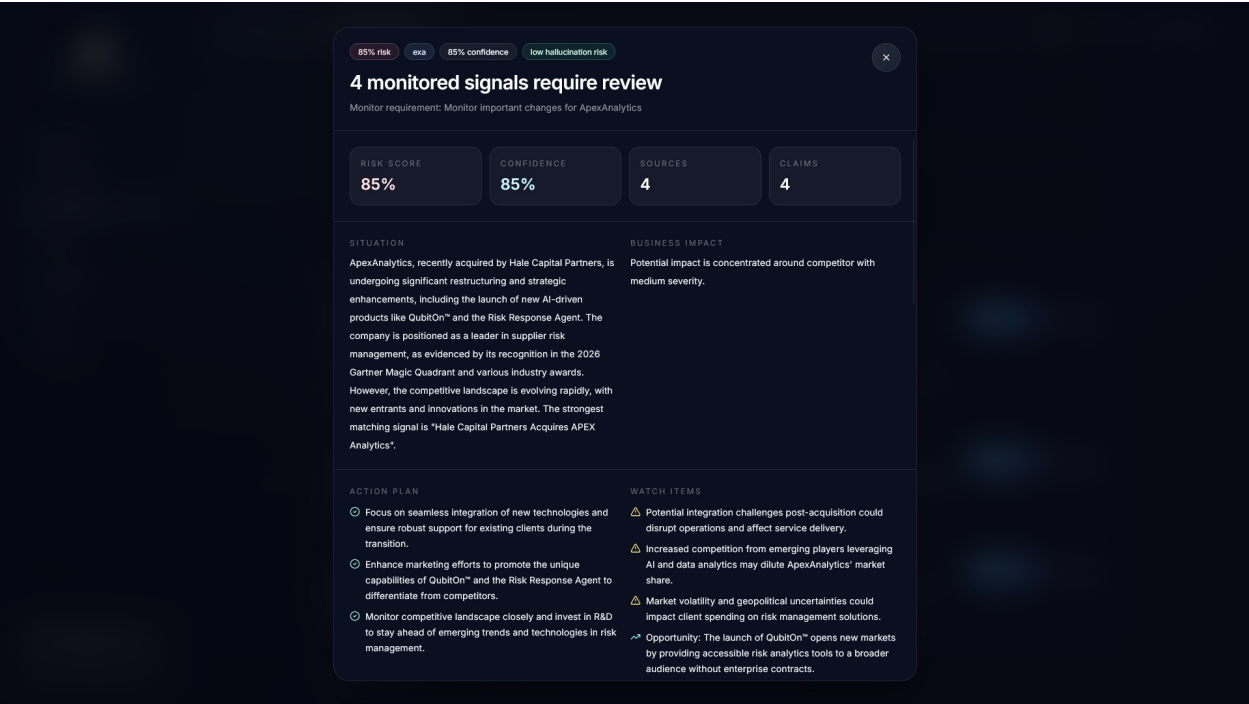
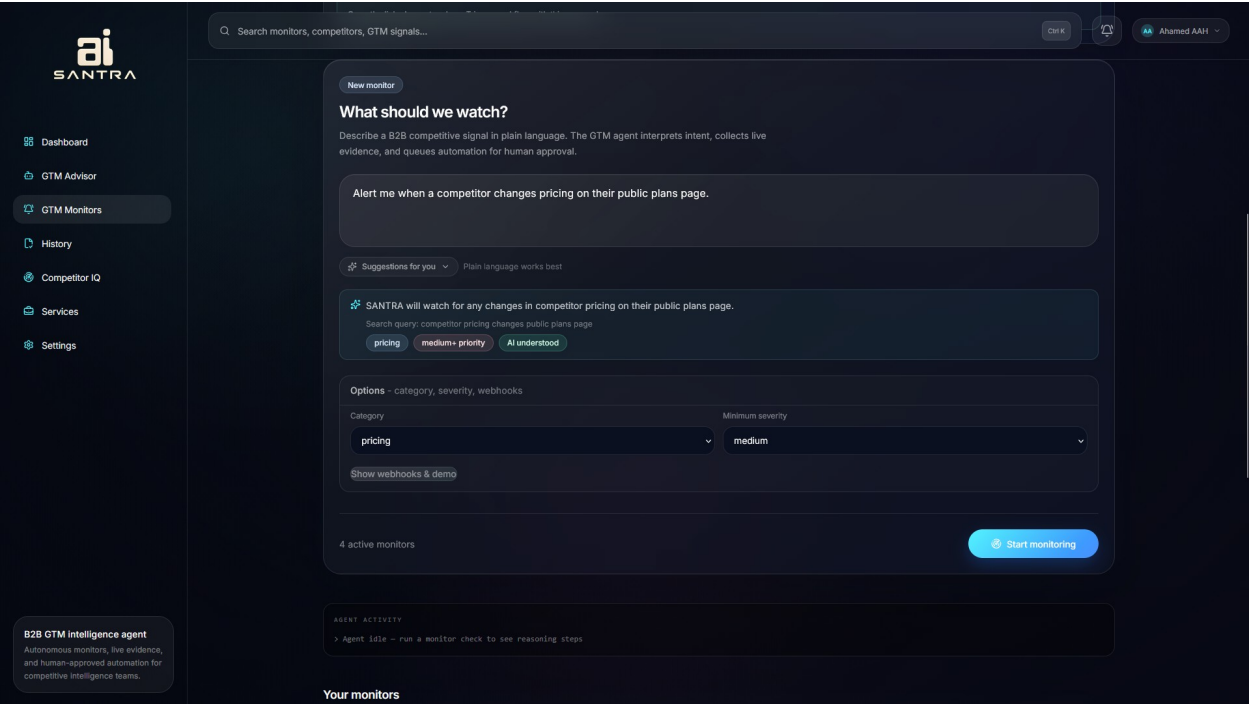
Add EXA_API_KEY, BAND_*, Supabase keys

npm run secrets:sync # AIML from vault

npm run dev # http://localhost:3001

Key env vars: EXA_API_KEY, BAND_AGENT_ID, BAND_API_KEY, Supabase URL/anon key, Bright Data tokens (optional), AIML/FEATHERLESS for LLM.







Action approval queue

The agent proposes CRM and automation actions. Nothing executes until a GTM owner approves.

Pending approval Monitor competitor pricing changes on public plans page

Review and approve CRM/automation trigger: 4 monitored signals require review

⌚ Approve

⌚ Dismiss

Approved — ready to execute

Review and approve CRM/automation trigger: 2 monitored signals require review

Open the linked report and use Trigger workflow with this approval.

Approved — ready to execute

Review and approve CRM/automation trigger: 4 monitored signals require review

Open the linked report and use Trigger workflow with this approval.

Intelligence ready

Live synthesis

Current Global Events Overview

Today's global landscape is marked by significant geopolitical tensions, economic fluctuations, and environmental challenges. Key events include escalating conflicts in Eastern Europe, economic instability in South America, and climate-related disasters in Asia.

RISK

72%

CONFIDENCE

85%

SIGNALS

4

SOURCES

4

Everything is generated - open a view below. The main page stays clean.

🔍 View full insight

🌐 Globe map (4)

🔍 Radar & reasoning

📊 Forecasts & graph

📁 Sources (4)

🗣️ Brief me (voice)

📅 Activity log

📄 Download brief

🔄 Scenario Pulse mode

